

EXPERIMENT 28: FIND THE BOUNDARY OF THE IMAGE USING CONVOLUTION KERNEL

Aim

To find and detect the boundaries of an image using a convolution kernel in Python and OpenCV.

Algorithm

1. Import OpenCV and NumPy libraries.
2. Read the input image.
3. Convert the image into grayscale.
4. Define a 3×3 convolution kernel for boundary detection.
5. Apply the convolution kernel using `cv2.filter2D()`.
6. Convert the filtered result into an 8-bit image.
7. Display and save the detected boundary image.

Convolution Kernel

```
-1 -1 -1  
-1  8 -1  
-1 -1 -1
```

Python Code

```
import cv2  
import numpy as np  
  
# Read the input image  
img = cv2.imread("input.jpg")  
  
if img is None:  
    print("Image not found")  
    exit()  
  
# Convert image to grayscale  
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

```
# Define convolution kernel for boundary detection
kernel = np.array([
    [-1, -1, -1],
    [-1, 8, -1],
    [-1, -1, -1]
], dtype=np.float32)

# Apply convolution
boundary = cv2.filter2D(gray, cv2.CV_32F, kernel)

# Convert result to 8-bit image
boundary = cv2.convertScaleAbs(boundary)

# Save output
cv2.imwrite("boundary_convolution_output.jpg", boundary)

# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Boundary Image", boundary)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the boundary of the image was successfully detected using a convolution kernel in OpenCV.